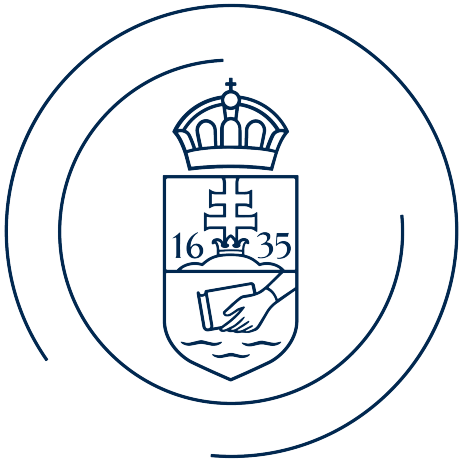


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Cybersecurity Basics

Mobile Security Fundamentals

Understanding Threats, Protections, and Best Practices

What Will You Learn?

Your Journey into Mobile Security

Foundations

- Why it matters
- Key challenges
- Real incidents

Threats

- Types of attacks
- Risks

Protection

- Authentication
- Best practices
- Security tools

Why Mobile Security Matters

The Stakes Are High

Your mobile device knows more about you than any other device you own!

What's At Risk:

- Banking credentials
- Health data
- Personal messages
- Location history
- Photos and videos



Unique Challenges in Mobile Security

Personal Data Hub	Concentrated storage of sensitive information
Always Connected	Constant connection to several networks of various types and security levels
Physical Mobility	Risk of loss, theft, shoulder surfing
Resource Limits	Limited processing power, battery life
App Ecosystem	Diverse sources, quality, and security

Mobile Security Threat Landscape

Critical Threats

- Social Engineering
  - Phishing attacks
  - Malicious apps
- Network Attacks
  - Man-in-the-middle
  - Rogue access points
- Malware & APTs
  - Banking trojans
  - Spyware

High-Risk Vectors

- Unsecured Wi-Fi networks
- Malicious apps
- Unpatched vulnerabilities
- Physical device access

Impact

- Financial loss
- Data theft
- Privacy breaches
- Identity theft

First Mobile Malware: Cabir (2004)

Key Characteristics

- First mobile worm targeting Symbian OS
- Spread via Bluetooth
- Displayed "Caribe" on startup

Impact

- Proof-of-concept for mobile malware
- UI disruption from repeated popups

Variants

Mabir

Enhanced version that could spread via:

- Bluetooth
- MMS
- .sis files

Historical Significance

Demonstrated mobile devices' vulnerability to malware, leading to increased focus on mobile security

Real-World Case Study: FluBot Banking Malware (2020–2022)

Attack Vector

- Fake SMS about package delivery/voicemail
- Malicious app installation request
- Requests accessibility permissions
- Spreads via contact list access

Impact

- Widespread infections (Spain, Finland, AU, JP)
- Banking credentials stolen
- Cryptocurrency theft
- Taken down by Europol (June 2022, 11 countries)

Key Warning Signs

- Link in SMS to install app
- Request for accessibility access

Real-World Case Study: Pegasus Spyware (2021–)



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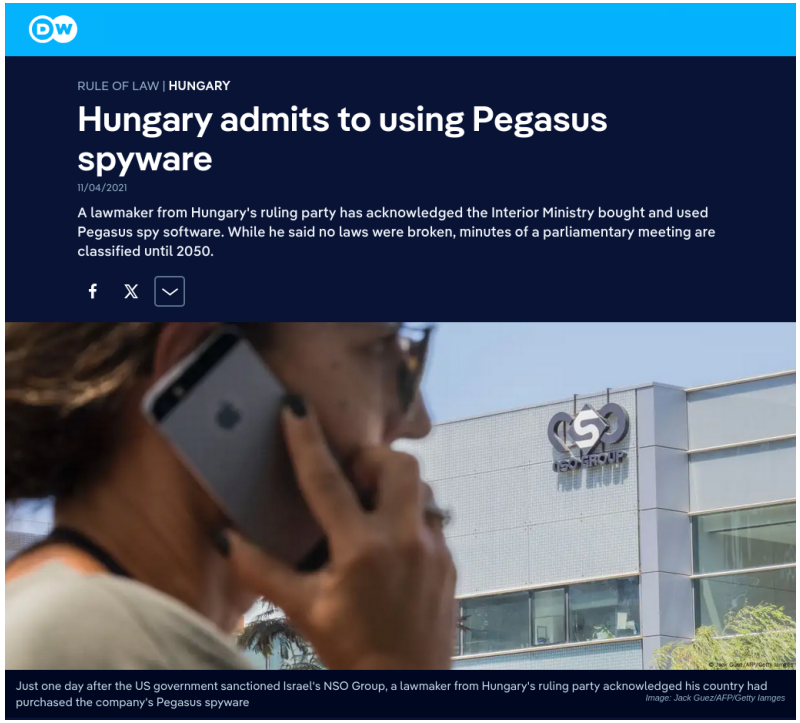
NEWS › CYBERSECURITY AND DATA PROTECTION

Brussels spyware crisis expands: Two MEPs hit in phone-hacking security breach

Lawmakers found malware infections on their phones, ringing alarm bells about confidential EU defense work.

- Zero-click exploitation
- Targeted journalists, activists, politicians
- Complete device compromise

Real-World Case Study: Pegasus Spyware (2021–)



- Multiple OS vulnerabilities
- Advanced persistence techniques

Case Study: DarkSword (2025–2026)

6-Stage Zero-Click Exploit Chain

- JavaScript on malicious / compromised websites
- No app install needed
- RCE → Sandbox Escape → Kernel Privilege Escalation



Threat Intelligence

The Proliferation of DarkSword: iOS Exploit Chain Adopted by Multiple Threat Actors

March 18, 2026

Google Threat Intelligence Group

Introduction

Google Threat Intelligence Group (GTIG) has identified a new iOS full-chain exploit that leveraged multiple zero-day vulnerabilities to fully compromise devices. Based on toolmarks in recovered payloads, we believe the exploit chain to be called DarkSword. Since at least November 2025, GTIG has observed multiple commercial surveillance vendors and suspected state-sponsored actors utilizing DarkSword in distinct campaigns. These threat actors have deployed the exploit chain against targets in Saudi Arabia, Turkey, Malaysia, and Ukraine.

DarkSword supports iOS versions 18.4 through 18.7 and utilizes six different vulnerabilities to deploy final-stage payloads. GTIG has identified three distinct malware families deployed following a successful DarkSword compromise: GHOSTBLADE, GHOSTKNIFE, and GHOSTSABER. The proliferation of this single exploit chain across disparate threat actors mirrors the previously discovered Coruna iOS exploit kit. Notably, UNC6353, a suspected Russian espionage group previously observed using Coruna, has recently incorporated DarkSword into their watering hole campaigns.

Spyware Democratization

Government-grade exploit kits increasingly shared among commercial vendors and criminal groups

Real-World Case Study: SparkCat (2025)

How It Worked

- Malicious SDK embedded in otherwise legitimate apps
- Published on **both** Google Play and the App Store
- Used on-device OCR to scan the user's screenshots

Scale

- Infected apps downloaded hundreds of thousands of times
- Discovered by Kaspersky, late 2024

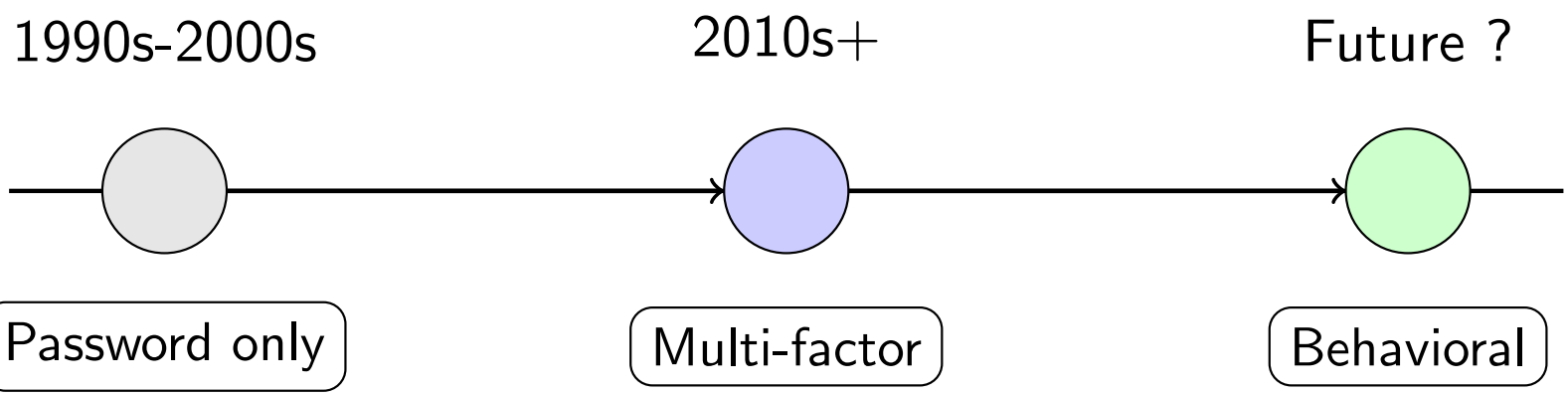
Key Lesson

Official app stores are not a guarantee of safety – malicious SDKs can slip past review

User Takeaway

Never store secrets unencrypted even as screenshots

Mobile Authentication Evolution



Past

- Text passwords
- Simple PINs
- Patterns

Present

- Two-factor auth
- Biometrics
- Hardware tokens
- Passkeys

Future

- Passwordless
- Continuous auth
- Context-aware

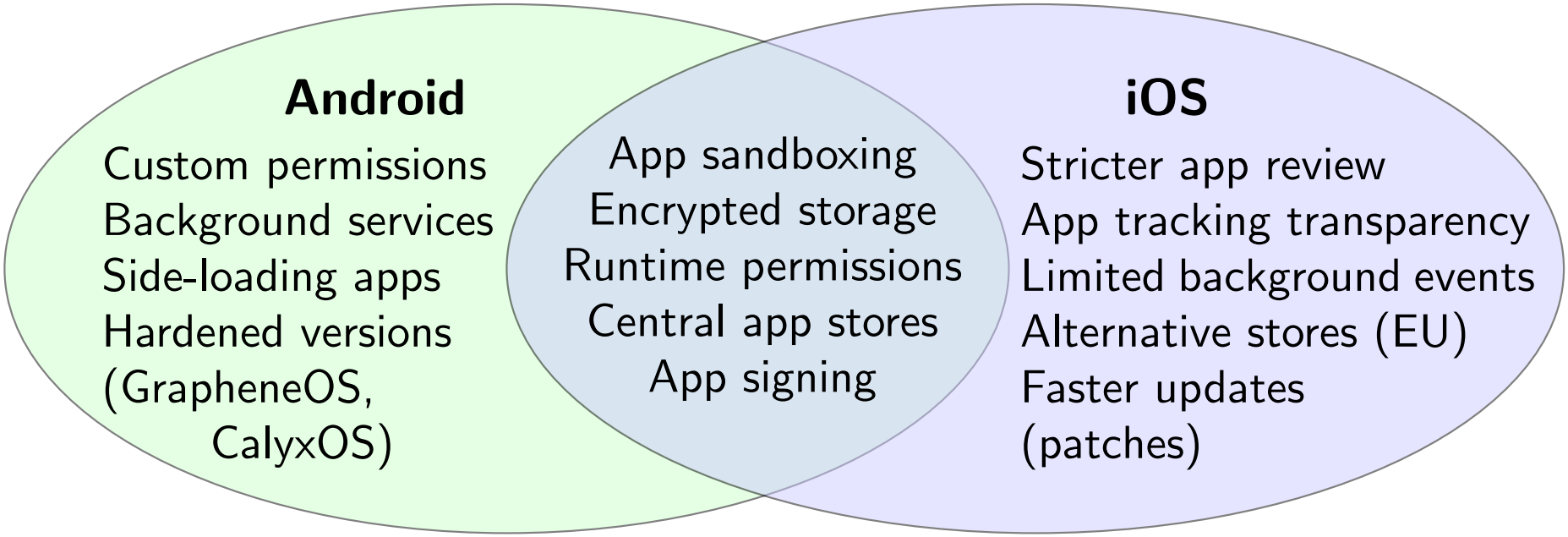
Security vs. Usability Trade-offs

The Eternal Balance

Strong security often comes at the cost of convenience, while maximum usability may compromise protection.

- **High Security**
  - Complex passwords (+security, -usability)
  - Multiple authentication factors (+security, -speed)
- **High Usability**
  - Biometric auth (+speed, -recovery options)
  - Longer session timeouts (+convenience, -security)

Security Features Compared



Developer Note

Always request minimum necessary permissions and explain why they're needed!

App Permissions Example: `AndroidManifest.xml`

Lists metadata about an app including permissions that could be requested

Example: **StreetComplete** (excerpt)

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android" xmlns:tools="h
  <uses-permission android:name="android.permission.INTERNET" />
  <uses-permission android:name="android.permission.ACCESS_NETWORK_STATE" />
  <uses-permission android:name="android.permission.ACCESS_FINE_LOCATION" />
  <uses-permission android:name="android.permission.ACCESS_COARSE_LOCATION" />
  <uses-feature android:name="android.hardware.location.gps" android:required="fals
  <uses-feature android:glEsVersion="0x00030000" android:required="true" />
  <application
    android:name="de.westnordost.streetcomplete.StreetCompleteApplication"
    android:allowBackup="false"
    android:icon="@mipmap/ic_launcher"
    android:label="@string/app_name">
    <activity android:name="de.westnordost.streetcomplete.screens.main.MainActivity
      <intent-filter>
        <action android:name="android.intent.action.MAIN" />
        <category android:name="android.intent.category.LAUNCHER" />
      </intent-filter>
    </activity>
  </application>
</manifest>
```

Mobile Network Security Landscape

Cellular	3G/4G/5G, encrypted by default
Wi-Fi	Various security standards (WEP $\rightarrow$ WPA3)
Bluetooth	Short-range, pairing-based security
NFC	Contactless payments, secure element
Satellite	Previously only custom frequencies and protocols, now Starlink can use LTE

Each connection type has its own security considerations.

Public Wi-Fi: Friend or Foe? (Reminder)

Dangers

- Man-in-the-middle attacks
- Evil twin networks
- Packet sniffing

Protections

- HTTPS – TLS certificate validation
- Disable auto-connect
- VPN

Understanding VPNs: Reality vs. Marketing

Actual Benefits

- Hides IP from visited sites
- Encrypts traffic from ISP
- Bypasses geo-restrictions
- Protection on public Wi-Fi

Limitations

- Sites may still track you (fingerprinting)
- VPN provider can see traffic
- Slower connection

Important Note

You're still trusting someone - instead of your ISP, you're trusting your VPN provider

- Consider: Free VPNs often monetize by selling user data
- HTTPS already encrypts most important traffic

Mobile Network Evolution and Security

2G (GSM)
- Weak crypto (A5/1, A5/2 – 40 bit)
- No network authentication
- Inter-core network (SS7) builds upon trust

3G (UMTS)
- Improved ciphers
- Same inter-core network
- IMSI catcher – privacy risk
- Downgrade attacks

4G (LTE)
- Strong cryptography (AES)
- New inter-core network (Diameter)
- Downgrade attacks still possible
- Attacks are harder, but not impossible (e.g. ReVoLTE, aLTEr)

5G
- Enhanced privacy features (depends on the provider)
- Phone (SUCI) tracking requires fake base station
- Downgrade attacks still possible

Core Mobile Security Features

Foundation of Mobile Security

Mobile security is built upon hardware, software, and user-facing features

Hardware Security

- Secure enclaves
- Biometric sensors
- Hardware-backed keys

Storage Security

- Full disk encryption
- File-based encryption
- AES-256 standards

Software Security

- Secure boot process
- App sandboxing
- App permissions
- Code signing
- Security updates

Key Principle

Security features work together in layers.

Biometric Authentication

Fingerprint

- Capacitive/ultrasonic sensors
- Under-display technology
- Template protection

Face Recognition

- 2D vs 3D sensing
- IR dot projection
- Liveness detection

Security Note

Biometric data never leaves the secure hardware - only template matching occurs

Encrypted Storage Implementation

Full Disk Encryption

- Entire device encrypted
- Single key for all data
- Boot-time protection

File-Based Encryption

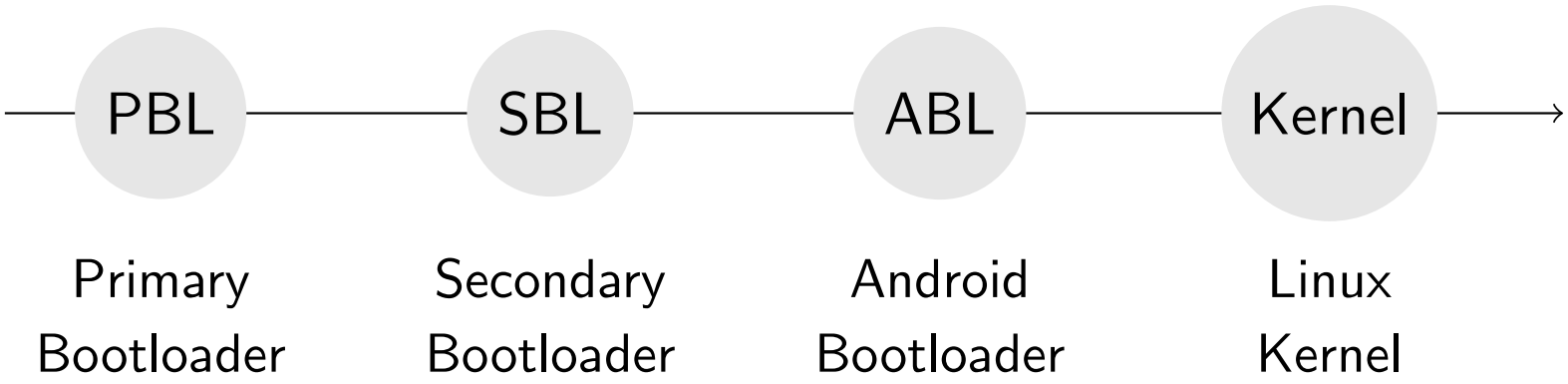
- Different keys per file
- Granular access control
- Better performance

Encryption Standards

Modern devices use AES-256 with hardware acceleration

Secure Boot Process

Android Boot Process



Chain of trust:

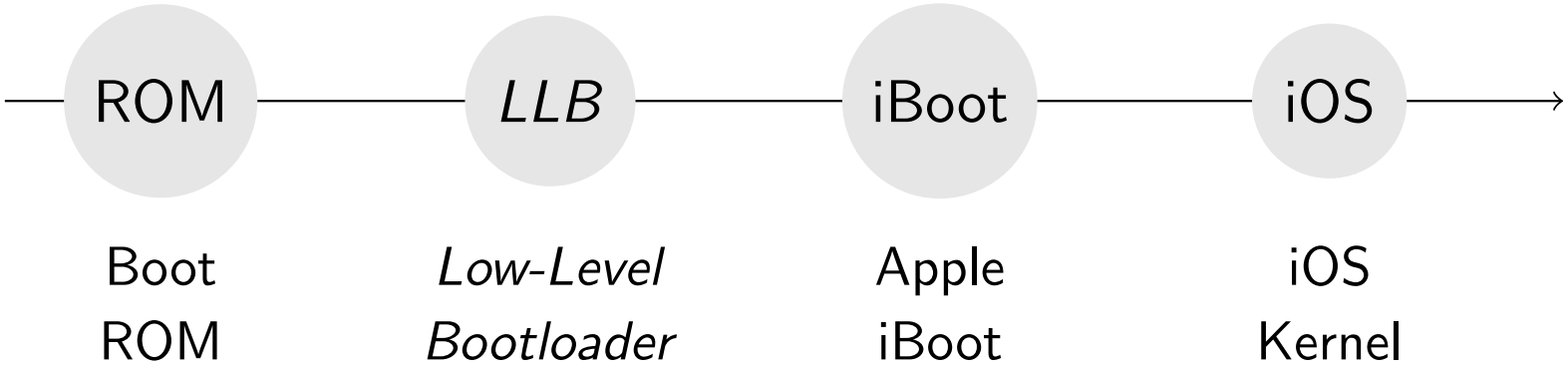
- Hardware root of trust
- Verify the signature of the next stage
- Continue to the next stage, repeat

Critical Function

Prevents boot-time malware and system modifications

Secure Boot Process

iOS Boot Process



Chain of trust:

- Hardware root of trust
- Verify the signature of the next stage
- Continue to the next stage, repeat

Critical Function

Prevents boot-time malware and system modifications

Check Your Device Settings

Essential Security Settings to Verify

- System and app updates
- App permissions and privacy controls
- Lost device protection
- Screen lock and biometric authentication
- Backups!

Hands-On

Take a moment to check these settings on your device. Which applications have access to your location, camera, or contacts? Is your data backed up?

Thank you for your attention!

Gergely Zsolt Kovács

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